AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	**ASHLESHA .S. BHOIR**
Roll No. :	**03**
Experiment No.:	**1**
Title of the Experiment:	**Perform Exploratory Statistical Analysis of a Real-World Dataset (Case Study)**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi Nagarhalli**

Signature :

Date :

Aim: To perform exploratory statistical analysis of the Pima Indians Diabetes Dataset by examining its structure, variable types, data quality, statistical characteristics, and important patterns.

Objectives

After completing this experiment, the student will be able to:

1. Apply exploratory data analysis techniques to understand the structure and characteristics of a real-world healthcare dataset.
2. Interpret the statistical characteristics, data quality issues, and important patterns identified during the analysis.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations, 8 input attributes, and one binary outcome variable.

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
2. Examine the dataset dimensions, column names, data types, and sample records.
3. Classify the variables as numerical, categorical, or binary.
4. Generate a statistical summary of the dataset.
5. Check for missing or invalid values.
6. Identify potential outliers using suitable statistical techniques and visualizations.
7. Generate appropriate visualizations to understand the data.
8. Record and interpret important observations obtained from the exploratory analysis.

Description of the Experiment

Exploratory Data Analysis (EDA) is the initial process of understanding a dataset before applying statistical inference or machine learning techniques.

The analysis follows: Raw Dataset, Data Inspection, Variable Identification, Data Quality, Analysis, Statistical Summary, Visualization and Interpretation

The Pima Diabetes Dataset is explored to understand its structure, data quality, variable distributions, and important characteristics before developing predictive models.

Detailed Description of Statistical techniques:

Descriptive Statistics

Statistical measures such as the following are used to summarize numerical variables:

- Mean
- Median
- Minimum
- Maximum

- Variance
- Standard Deviation

Missing Value Analysis

The dataset is examined for missing or invalid values.

For example, zero values in certain medical attributes such as **Glucose, BloodPressure, SkinThickness, Insulin, and BMI** may require investigation because a zero value may represent missing or invalid data rather than a genuine medical measurement.

Outlier Analysis

Potentially unusual observations are identified using statistical summaries and visualizations such as boxplots.

An outlier should be investigated before deciding whether it should be removed or retained.

Data Visualization

Suitable visualizations are generated to understand:

- Variable distributions
- Frequency of diabetes outcomes
- Relationships between variables
- Potential outliers

Examples include:

- Histograms
- Boxplots
- Bar charts
- Scatter plots

Implementation

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[1] ✓ 4.2s Python

```python
df = pd.read_csv("diabetes.csv")
```

[2] ✓ 0.0s Python

```python
df.head()
```

[ ] Python

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulir
0	6	148	72	35	(
1	1	85	66	29	(
2	8	183	64	0	(
3	1	89	66	23	94
4	0	137	40	35	16

```python
df.shape
```

Python

```
(768, 9)
```

```python
df.columns
```

Python

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThic
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome']
      dtype='str')
```

```python
df.dtypes
```

Python

```
Pregnancies                   int64
Glucose                       int64
BloodPressure                 int64
SkinThickness                 int64
Insulin                       int64
BMI                         float64
DiabetesPedigreeFunction    float64
Age                           int64
Outcome                       int64
dtype: object
```

```python
df.info()
```

Python

```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

```python
df.describe()
```

Python

	Pregnancies	Glucose	BloodPressure	SkinThickness
count	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458
std	3.369578	31.972618	19.355807	15.952218
min	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000
75%	6.000000	140.250000	80.000000	32.000000
max	17.000000	199.000000	122.000000	99.000000

```python
df.isnull().sum()
```

Python

```
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

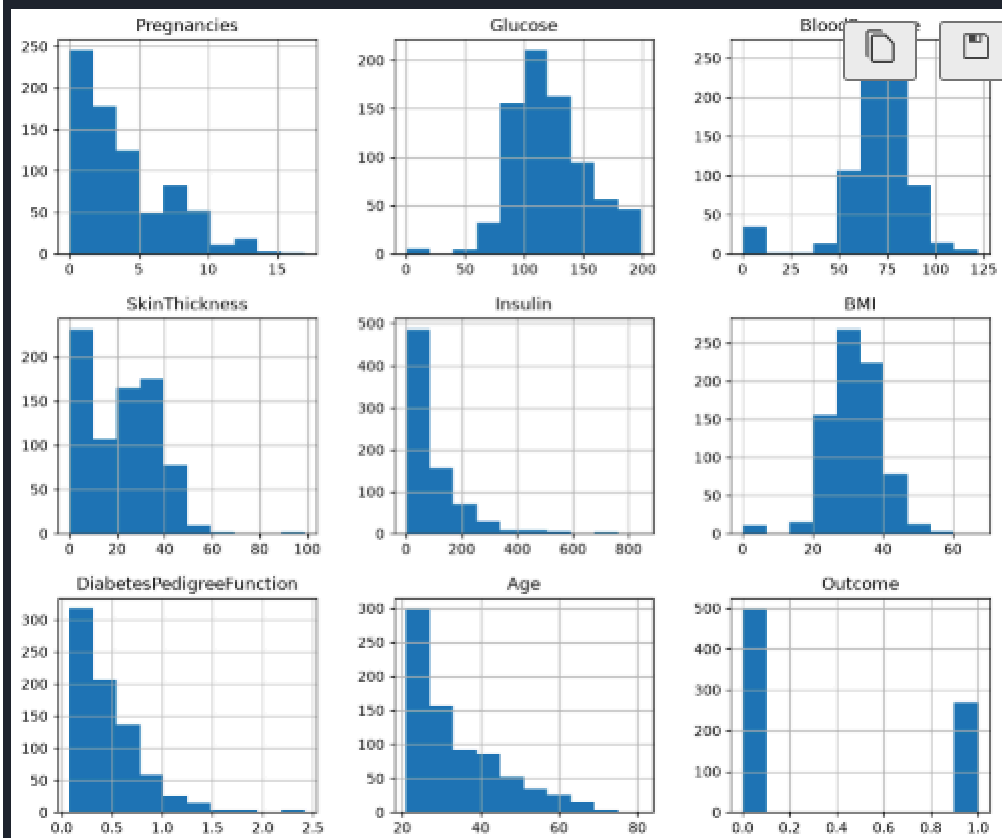
```python
(df == 0).sum()
```

Python

```
Pregnancies                 111
Glucose                       5
BloodPressure                35
SkinThickness               227
Insulin                     374
BMI                          11
DiabetesPedigreeFunction      0
Age                           0
Outcome                     500
dtype: int64
```

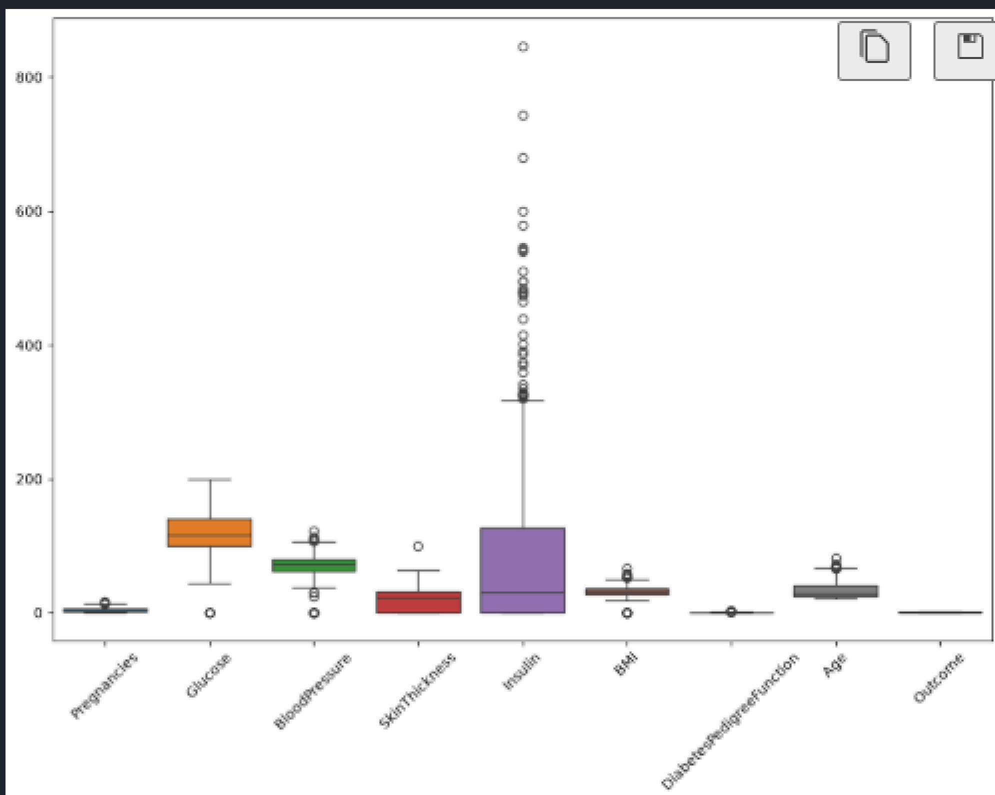
```
df.hist(figsize=(12,10))
plt.show()
```

Python



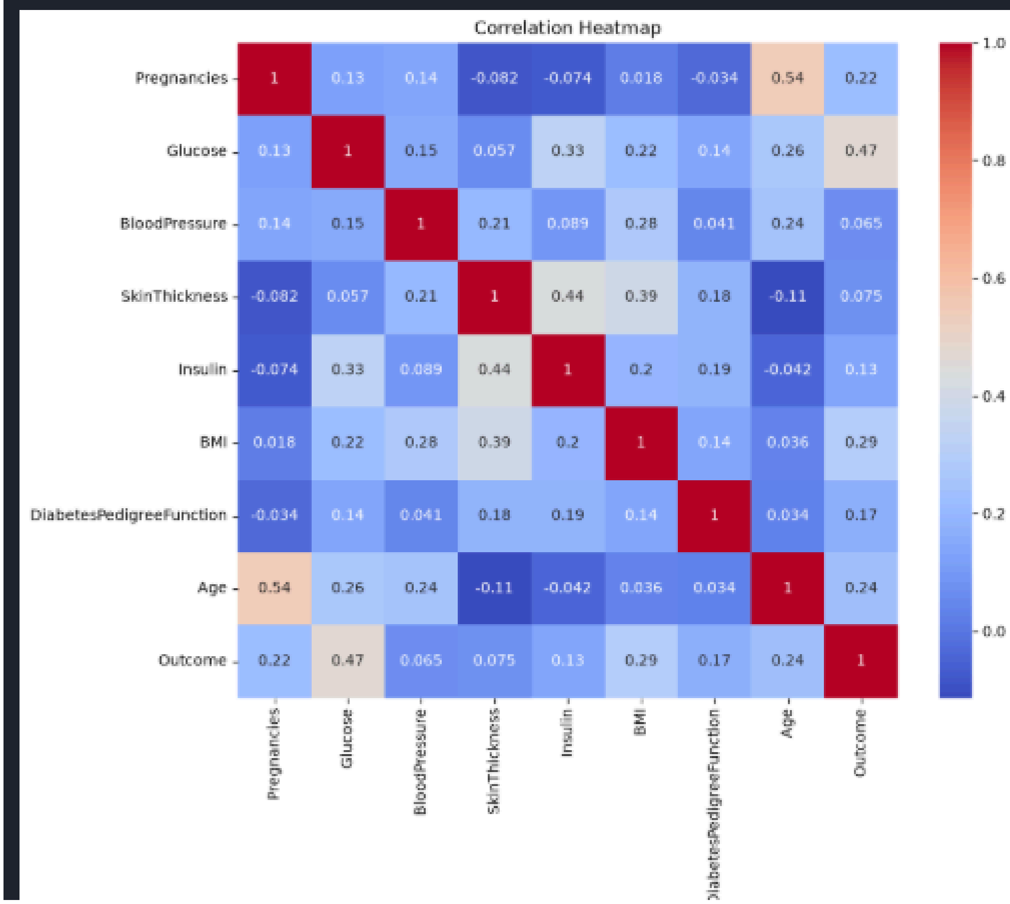
```python
plt.figure(figsize=(12,8))
sns.boxplot(data=df)
plt.xticks(rotation=45)
plt.show()
```

Python



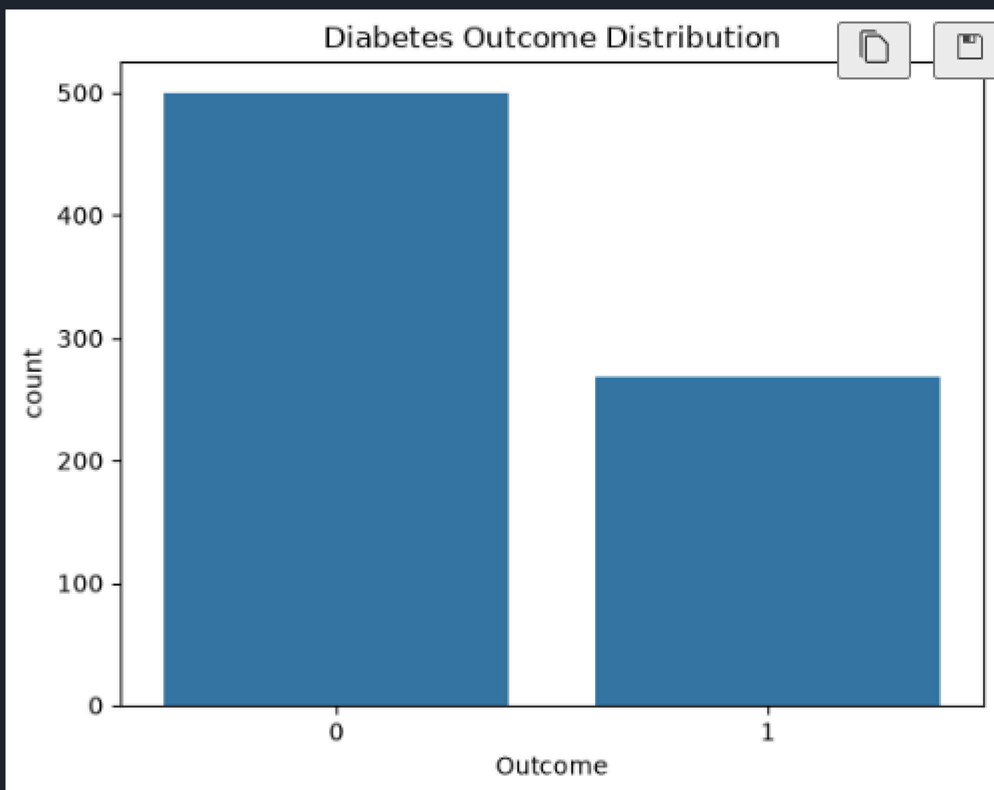
```python
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Python



```python
sns.countplot(x="Outcome", data=df)
plt.title("Diabetes Outcome Distribution")
plt.show()
```

Python



```python
sns.scatterplot(x="Glucose", y="BMI", hue="Outcome", da
plt.title("Glucose vs BMI")
plt.show()
```

Python

Conclusion

Exploratory statistical analysis was performed on the Pima Indians Diabetes Dataset to understand its structure, variable characteristics, data quality, and important patterns. The analysis provides the foundation for further statistical analysis and machine learning model development.

Questions to be Answered by Students

1. Which variables in the dataset contain zero or potentially invalid values that require investigation?

   ANS: The following variables contain zero values that are physiologically impossible and should be treated as missing or invalid values:

   - Glucose
   - BloodPressure
   - SkinThickness
   - Insulin
   - BMI

2. Which variable contains potential outliers based on the analysis?

   ANS: The Insulin variable contains the most significant outliers, as indicated by boxplot analysis. Other variables such as BMI, SkinThickness, and DiabetesPedigreeFunction also exhibit some outliers, but Insulin has the highest variability and extreme values.

3. What are the important patterns observed between the input variables and the Outcome variable?

   ANS: Glucose shows the strongest positive relationship with the Outcome; individuals with higher glucose levels are more likely to have diabetes.

   Higher BMI values are associated with a greater likelihood of diabetes.

   Older individuals (Age) tend to have a higher probability of being diabetic.

Women with a higher number of Pregnancies generally show a higher incidence of diabetes.

DiabetesPedigreeFunction has a moderate positive association with diabetes, indicating the influence of family history.

BloodPressure, SkinThickness, and Insulin exhibit comparatively weaker relationships with the Outcome.

4. State **two important insights** obtained from the exploratory analysis.

   ANS: Glucose is the most influential predictor of diabetes, with diabetic patients showing considerably higher glucose levels than non-diabetic patients.

   The dataset contains several invalid zero values in medical measurements (Glucose, BloodPressure, SkinThickness, Insulin, and BMI), which should be handled through data preprocessing before building machine learning models to improve prediction accuracy.